

phone number, the serial number of the unit and the target telephone number. The BTS forwards this data to the MTSO. The MTSO in turn, validates the request and makes a physical connection through the switching network, and instructs the BTS and the mobile phone user to move over to an unused voice channel, to allow the conversation to begin. Often the request may be made to the PSTN, to make a connection to a number within the PSTN. Since these events typically occur within a few seconds, they are not noticeable to the user.

Once a call is in progress, the MTSO governing the call adjusts the transmitted power of the phone through the BTS and changes the channel of the mobile phone and the BTS in an effort to maintain call quality as the user moves within the range of each BTS. Since the user is mobile, the user may tend to weave in and out of the range of a BTS—when that happens, a sophisticated switching technique known as a handoff enables the call to proceed as the user moves to the range of another BTS and is handed-off into another unused voice channel. This ensures smooth and uninterrupted service for the user.

The fine print

The advantage of CDMA over the other technologies is that there is no specified absolute limit on the number of users on the network. Rather, the system performance very gradually degrades for all the users as the number of users goes up, and improves as the number of users is decreased. Of course, if all the users on the network decide to simultaneously clog up the network, it will tend to go down. Also there is the point of spectral efficiency. CDMA makes maximum efficient use of the available spectrum allotted to it. Moreover, since the message signal is spread over a wide spectrum, the resultant signal is more powerful and offsets the effects of small scale fading. These features translate to longer battery lives and talk time for CDMA users. As with every new technology, CDMA isn't without its own set of problems. Most CDMA services don't use a SIM (Subscriber Identity Module) that is typically used in GSM phone services. Instead the carrier generally programs the phones. This means that if you would want to shift carriers at a later



There are 800 million GSM service users and 150 million CDMA service users globally

It's not all Hunky-Dory

In a country of 1 billion potential consumers, the carrier war appears to have just begun. The bone of contention here is the so-called hybrid WLL services that are now being rolled out by Reliance Info-comm and the Tata group. Backed by the CDMA technology, customers will have their cake and can eat it too. With super-low charges being the main attraction to shifting your carrier, the older GSM-based carriers are fighting back crying foul over the charges. The cases have been filed and are being argued out in the courts. What this essentially means for the average Indian is that till the GSM, CDMA, as well as TRAI (the Telecom Regulatory Authority of India), get down to resolving their differences, we are stuck with incompatible or

partially compatible services.

To add further fuel to the fire, there appears to be a new technical problem regarding the co-existence of GSM and CDMA. GSM operators point out that CDMA transmissions now affect the quality GSM services. The controversy has arisen because the high-powered CDMA transmitters are beaming out pretty close to the frequencies of GSM services that are comparatively low powered. This results in call drops and poor receptions and not to mention, overheating of GSM mobile phones. The solution appears to be in the usage of frequency filters that prevent such leakage—just another petty squabble or a real problem of incompatibility? Only time will tell.

date, it would be upto your carrier to have the phone reprogrammed. Another point here is that the CDMA consumer base is comparatively less than the GSM base. This means that your CDMA service provider today may not yet be able to provide you the kind of roaming services that we take for granted

in GSM networks—but then again, with some of the biggest names in the industry pushing for CDMA in India, it may soon be possible, at least within India. This is not due to any technical limitation of CDMA—it's all about the geo-technical politics of the mobile world. Overall, CDMA

technologies are easier to implement and maintain. Additionally, one rather controversial point is that the running costs of CDMA are lower than those of competing technologies. And as far as the consumer goes, CDMA in its current form is relatively future-proofed.

CDMA and 3G

Virtually all of today's mobile networks use the second generation, 2G technologies, or better. Although it sounds rather exciting, this is nowhere near what is being promised in the future. Later in this decade, we could be drowning in world of incredible multimedia content. The idea is that as most mobile communication networks move beyond to the promised manna of 3G, they will unveil services that could be the literal

equivalent of having broadband on your handheld. Look forward to seeing full-blown colour videos, crisp polyphonic audio and live videophony. It's already happening—look at Japan's NTT-DoCoMo, and South Korea. With the step up from CDMAOne to CDMA2000 1x already happening in places around the world, we could soon be seeing data speeds of up to 307 Kbps—and if we look just one step beyond, at the second phase of 3G, with technologies such as CDMA2000 1xEV DO (Evolution, Data Only), we could be looking at speeds of up to 2.4 Mbps!

What would this bunch of airborne goodies get us? If all the carriers had their way, with 3G, every country on the planet would turn out to be like South Korea—all of us would be experiencing broadband Web access on our mobile devices, not on our desktops. The mobile phone isn't just a phone anymore; it is evolving into an entertainment centre. With 3G, the carriers are hoping that we will be playing multiplayer games, querying Google, sending and receiving e-mail, and sending off full-colour, full-audio videos to just about everyone all over the planet!

It may well be possible. Remember that within five years of commercial deployment of CDMA services, the first 3G networks had appeared. The entrance of WLL and CDMA services in this country could well be the precursor to putting India online. 

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